

MicroHasTEE: Bare-Metal Haskell for Type-Level Peripheral Ownership on TrustZone-M

A Framework Realized on STM32 Nucleo Boards

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Abstract

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1 Introduction

Arm TrustZone for Armv8-M, or TrustZone-M, partitions a microcontroller into hardware-isolated Secure and Non-secure states. It also lets Secure firmware expose explicitly declared entry points to Non-secure code. TrustZone-M does not, however, determine which state owns each peripheral, GPIO pin, or SRAM region. Vendors implement these assignments through device-specific security hardware. On the STM32U5 Nucleo board used in this work, memory-mapped security registers, including those of ST's Global TrustZone Controller (GTZC), encode the security attribution of peripherals and SRAM. Transferring a resource between states therefore requires developers to update this STM32-specific configuration and keep initialization, interrupt routing, and access permissions consistent with it.

Conventional STM32 development turns this configuration into a coordination problem between two separately built firmware images. Each image can type-check while relying on assumptions about resource ownership that conflict with those of the other image. Secure code may continue to use a UART after the firmware has assigned it to the Non-secure state, or Non-secure code may assume access to a GPIO pin that the Secure configuration never released. The hardware enforces the configured boundary at run time, but it cannot determine whether the two applications agree

about that boundary. Developers therefore discover many configuration mismatches only when they execute the firmware on the target device.

MicroHasTEE instead treats TrustZone firmware as a multiparty Haskell program whose participants are the Secure and Non-secure applications. Although these participants execute as separate firmware images, the programmer implements them simultaneously in one source-level program. This program describes the computations performed by each participant, the resources available to them, the transfer of resources between them, and their permitted cross-boundary calls. MicroHs [1] compiles the shared program separately for each TrustZone state. Each resulting image contains the MicroHs runtime and executes compiled Haskell application code directly on the microcontroller without an operating system or RTOS. A small C layer provides startup and raw peripheral access, while the application logic, resource partitioning, and cross-state interactions are expressed and executed as Haskell computations. To the best of our knowledge, MicroHasTEE is the first framework to execute Haskell in both states of a bare-metal TrustZone-M microcontroller and to use a single typed Haskell program to coordinate their resource assignments and cross-state interactions.

This shared description lets MicroHasTEE check assumptions that would remain unrelated in separately written applications. The type checker verifies that each participant uses only the resources currently available to it and that both participants agree on resource transfers. It also checks cross-boundary calls against the Secure functions exposed to the Non-secure participant. Both images are therefore checked against one account of their shared hardware and interface, rather than only being checked in isolation. A disagreement becomes a compile-time error instead of a fault observed after deployment.

These guarantees apply to operations expressed through MicroHasTEE. The framework cannot account for arbitrary C or FFI code that accesses a peripheral directly or modifies security registers outside its interface. The MicroHasTEE programming model is not specific to the STM32U5. It can target other TrustZone-M microcontrollers with sufficient flash and SRAM, provided that a backend realizes the required memory, peripheral, and interrupt attribution mechanisms. The implementation evaluated in this work uses the STM32U5-specific security hardware.

This work makes three contributions:

- **A multiparty programming model for TrustZone-M firmware.** MicroHasTEE lets developers implement Secure and Non-secure applications together as participants in one source-level program and then compile that program into separate firmware images.

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- **Static agreement about resources and cross-boundary interfaces.** MicroHasTEE checks each participant's resource use against a shared account of resource authority, transfers, and permitted calls, allowing it to reject inconsistencies before the firmware reaches the target device.
- **Bare-metal Haskell in both TrustZone-M states.** The prototype embeds the MicroHs runtime in each firmware image and executes compiled Haskell application code in both states without an operating system or RTOS. It connects the high-level shared resource description to the STM32 security configuration and hand-written peripheral drivers.

2 Conventional Bare-Metal TrustZone-M Development

Constructing a conventional bare-metal TrustZone-M system requires the programmer to express one security partition through several independent hardware and software mechanisms. This section first introduces the mechanisms defined by Armv8-M and then explains how STM32 microcontrollers supplement them with device-specific controls for memory, peripherals, and interrupts. Together, these mechanisms form the low-level programming model that MicroHasTEE seeks to make safer.

2.1 Memory Partitioning

A microcontroller contains fixed amounts of flash memory and SRAM, each of which may comprise several physical banks. A linker script assigns a firmware image to address ranges and places its vector table, code, static data, and stack within them. Figure 1(a) shows this layout for a conventional bare-metal application. Because Secure and Non-secure firmware are built as separate images, each image requires its own linker script. The Secure linker script must also reserve a Non-secure Callable region for the gateway veneers used to enter Secure code.

TrustZone-M allows developers to partition memory into Secure and Non-secure regions. Our implementation assigns one primary flash range and one primary SRAM range to each security state and reserves a Non-secure Callable subregion within Secure flash. The MCU's alignment and block-size requirements constrain the boundaries of these regions. As fig. 1(b) shows, the two firmware images occupy disjoint regions of the same physical flash and SRAM.

The linker scripts describe this intended layout, but they do not enforce its security attributes. Secure startup code must configure the architectural and STM32-specific attribution mechanisms to enforce the same partition. The two images can therefore link successfully even when the linker and hardware configurations disagree. Such a mismatch typically triggers a fault only when firmware accesses the affected region.

On the STM32U5, ST's *Implementation Defined Attribution Unit* (IDAU) assigns fixed security attributes to address ranges. The Cortex-M33 supplements this fixed mapping with a programmable *Security Attribution Unit* (SAU). For each CPU access, the processor combines the IDAU and SAU results and selects the more restrictive attribute. The processor orders these attributes as Secure > Secure, Non-secure Callable > Non-secure.

The IDAU and SAU classify only transactions issued by the CPU, while other bus masters can access SRAM without passing through

them. ST therefore protects each SRAM bank at its memory interface with a block-based memory protection controller (MPCBB). Each MPCBB assigns Secure or Non-secure attributes to individual SRAM blocks. The SAU/IDAU and MPCBB configurations must assign compatible attributes to every SRAM region used by the firmware.

2.2 Peripheral Attribution with the GTZC

Armv8-M defines the Secure and Non-secure states, memory attribution, and the mechanisms for transferring control between the two states. STM32 microcontrollers supplement these architectural mechanisms with controls that assign security attributes to memory-mapped peripherals. The *Global TrustZone Controller* (GTZC) includes a *TrustZone Security Controller* (TZSC), which configures the security attribution of supported peripherals. Although the GTZC provides additional protection and monitoring mechanisms, we focus here on the TZSC's role in peripheral attribution.

Each peripheral controlled by the TZSC has a corresponding bit in a security configuration register. A Secure attribution blocks Non-secure transactions, whereas a Non-secure attribution permits them. A Non-secure attribution therefore does not give the Non-secure state exclusive access to the peripheral.

Some STM32 peripherals provide finer-grained attribution through their own security registers. For example, each GPIO port controls up to 16 pins and provides a security configuration bit for each implemented pin. Secure software can therefore retain Secure access to selected pins while permitting Non-secure access to other pins on the same port. Only Secure software can modify these security attributes.

2.3 Interrupt Attribution

Peripheral attribution does not by itself determine where interrupts from that peripheral execute. Armv8-M assigns each external interrupt a target security state through the Interrupt Target Non-secure State (ITNS) registers in the Nested Vectored Interrupt Controller (NVIC). Secure software configures this target, and the processor uses either the Secure or the Non-secure vector table when it takes the interrupt.

STM32 devices may also attribute the interrupt source independently of its NVIC target. For example, moving a button input to the Non-secure application requires Secure firmware to make the GPIO pin Non-secure, configure the corresponding EXTI line and mark it Non-secure, and target its NVIC interrupt to the Non-secure state. The Non-secure image must also contain a handler in the corresponding slot of its vector table. These settings must agree before the interrupt is enabled. Releasing the GPIO pin alone is therefore insufficient: if the EXTI attribution, NVIC target, or vector entry disagrees, the event may be delivered to the wrong state or to a handler that cannot service it.

2.4 Non-secure Callable Functions and Gateway Veneers

The Secure and Non-secure firmware images occupy disjoint regions of flash and SRAM. This isolation is asymmetric: Secure software can access Non-secure resources, whereas Non-secure

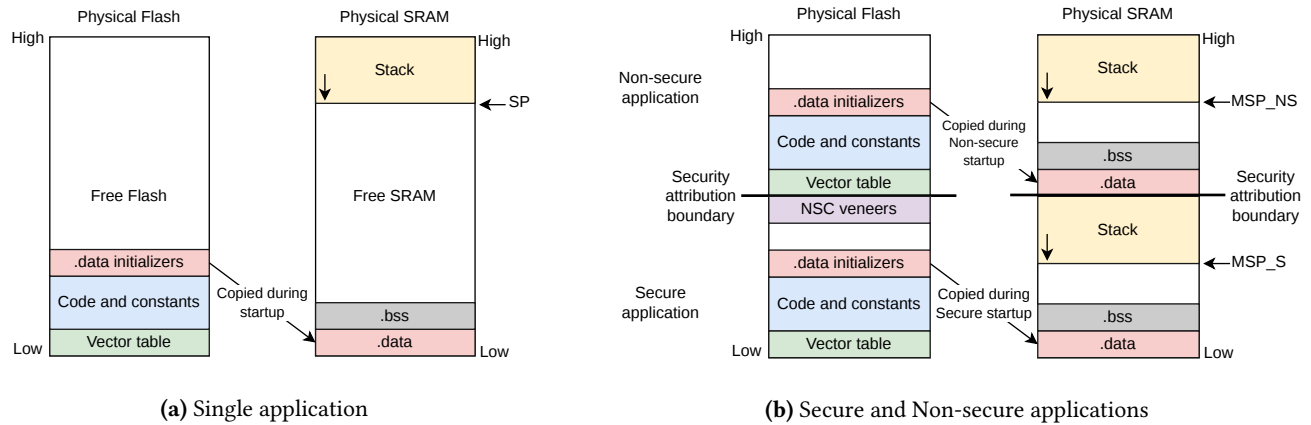


Figure 1: Physical flash and SRAM layouts for a single bare-metal application and two TrustZone-M applications. The layouts are not drawn to scale.

software cannot directly access Secure memory or peripherals. Secure firmware can nevertheless expose selected entry functions to Non-secure callers without exposing its remaining code and data.

Secure firmware declares an entry function using the CMSE `cmse_nonsecure_entry` attribute. The compiler marks the function as a Secure entry point, and the Secure linker creates a corresponding secure gateway veneer in a Non-secure Callable region. The veneer consists of an SG instruction followed by a branch to the entry function in Secure memory. Non-secure code calls the exported gateway symbol as an ordinary function. When the processor executes SG at a valid gateway, it changes to the Secure state and continues to the entry function. The compiler-generated epilogue clears register state as required and executes BXNS to return directly to the Non-secure caller.

Below is a simple function defined by the Secure application, with an attribute that marks it as a NSC function.

```
1 int __attribute__((cmse_nonsecure_entry))
2 add10(int value) {
3     return value + 10;
4 }
```

The Non-secure application declares the function and calls it like an ordinary external function. The linker will resolve it when the firmware is built.

```
1 extern int add10(int value);
2
3 int myadd(void) {
4     return add10(5);
5 }
```

This declaration is sufficient to compile the Non-secure source, but it is not sufficient to link the Non-secure image. The address associated with `add10` is the address of the gateway veneer created when the Secure image is linked. With the GNU toolchain, the Secure link uses `-cmse-implib` and `-out-implib` to emit a CMSE import library containing the exported symbols and their veneer addresses. The Non-secure linker must then consume this generated library.

The simplified build dependency is therefore:

```
1 secure.elf:
2 $(CC) ... -Wl,--cmse-implib,\
3     --out-implib=secure_cmse_import.lib
4
5 nonsecure.elf: secure.elf
6 $(CC) ... secure_cmse_import.lib
```

The two images are linked separately, but they cannot be built independently. The Secure link must run first, and any change to the exposed interface or veneer placement requires the import library and Non-secure image to be regenerated together. Moreover, the C linker resolves `add10` by symbol name and address; it does not verify that the independently compiled declarations on the two sides have matching types.

The control flow of this call from Non-secure to Secure code is shown in fig. 2.

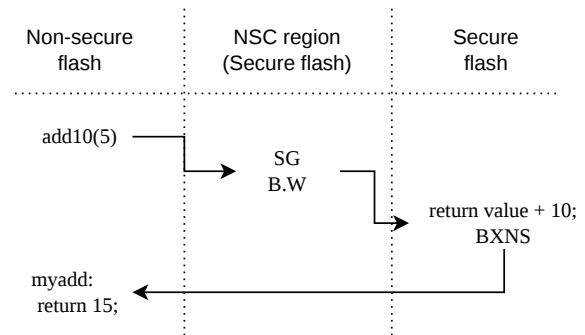


Figure 2: Control flow of a Non-secure call to a Secure entry function. The SG instruction in the NSC gateway veneer changes the processor to Secure state, after which B.W branches to the Secure function body. The function returns directly to its Non-secure caller with BXNS.

A Secure entry function must treat pointer arguments received from Non-secure code as untrusted. Before dereferencing a pointer, it must verify that the entire referenced range lies in Non-secure memory, has the permissions required by the operation, and does not wrap around the address space when its size is added. After validating an input range, Secure code copies its contents into Secure memory and performs subsequent checks and computations on that copy, preventing later modifications to the original buffer from changing the value being processed.

This section describes several relatively small mechanisms. The difficulty arises when they need to be used together to agree on a security policy. The simple act of allowing a Non-secure button handler to call a Secure service involves peripheral and interrupt attribution, a Non-secure vector entry, matching declarations in two programs, a gateway veneer, and an ordered build in which the Secure link produces input to the Non-secure link. Each mechanism may by itself look correct, while some other part of the system disagrees with the policy.

3 Threat Model and Scope

MicroHasTEE addresses two distinct concerns. Its type system detects configuration inconsistencies introduced by a trusted but fallible developer, while TrustZone-M provides run-time isolation from compromised Non-secure firmware.

We assume that matching firmware images are initially generated from the same MicroHasTEE program and deployed together. The adversary may then execute arbitrary Non-secure code, read and modify Non-secure memory, use Non-secure-attributed peripherals, and invoke NSC gateways with arbitrary pointers and messages. Any service registered as callable is available to the entire Non-secure domain, because callable handles are not authorization capabilities. We assume that the adversary cannot modify Secure firmware, change security attribution, directly access Secure resources, or enter Secure code except through designated NSC gateways. Our run-time objective is to preserve the confidentiality and integrity of Secure code, data, and peripherals, except for effects intentionally exposed by registered services.

MicroHasTEE uses a separate fault model for the developer. For programs expressed through its public interface, the type system checks resource transfers, resource use, callback placement, and the registration of callable services. After `lock_configuration`, well-typed code cannot perform further attribution-changing operations through that interface. These guarantees do not apply to raw C, foreign-function calls, or internal modules, and the type-level ledgers express software authority rather than exclusive hardware access.

The guarantees assume that both builds execute corresponding setup operations in the same order and that MicroHasTEE, its hardware backend, the Secure gateway, the MicroHs toolchain and runtime, and the underlying hardware are correct. Malformed gateway messages are not covered by the static types but remain within the run-time threat model. The gateway must validate and copy Non-secure input before trusted deserialization code processes it. We exclude physical attacks, fault injection, side channels, invasive debugging, firmware rollback, denial of service, and faults within the trusted computing base.

4 MicroHasTEE

MicroHasTEE is a multitier programming framework in which the Secure and Non-secure applications are participants in one source-level program. The program describes each participant's computations, the resources assigned to it, and the interactions permitted between the participants. The build compiles this program twice, once for each TrustZone domain. In the Secure build, `Secure` computations execute while `Nonsecure` computations are represented by placeholders; the Non-secure build reverses these roles. The two builds select different implementations of the same MicroHasTEE interface, so they type-check the same setup sequence and derive corresponding views of the shared program. The program's setup phase assigns security attribution, configures peripherals, registers callbacks and Secure services, and starts the Non-secure application.

MicroHasTEE represents resource ownership with a pair of type-level capability ledgers, one for each participant. The `Setup` monad describes a transition from one pair of ledgers to another.

```
1 data Setup nsPre sPre nsPost sPost a
```

The parameters `nsPre` and `sPre` describe the Non-secure and Secure ledgers before the setup computation. The parameters `nsPost` and `sPost` describe the corresponding ledgers afterward. The final parameter, `a`, denotes the value returned by the computation. Each public attribution operation updates the hardware configuration and performs the corresponding transition between ledger pairs. Assuming a correct hardware backend and no access through raw C, FFI, or internal interfaces, the final pair therefore describes the attribution established during setup. Subsequent MicroHasTEE operations require evidence that their resource occurs in the executing participant's ledger, so the compiler rejects operations for which that participant lacks authority.

The configuration phase always begins with MicroHasTEE program being *open to configuration*.¹ This is encoded by the capability ledger for the Secure domain carrying a `Unlocked` entry. Consider the program below

```
1 configure :: Setup '[ ] '[Unlocked] '[UART] '[Locked] ()
2 configure = do
3   uart <- get_console
4   -- configuration of the actual UART device omitted
5   tzsc_release_periph uart
6   lock_configuration
```

The type of `configure` says that initially, the Non-secure application have no capabilities and the Secure capabilities indicate that MicroHasTEE is open for configuration. The final ledgers assign the UART to the Non-secure participant and leave the Secure participant with no tracked peripheral capabilities. The `Locked` entry records that further configuration is prohibited. The intermediate types show the history of what happens with the type of the function

```
1 -- Initially ns = [ ], s = [Unlocked]
```

¹For readability, the listings in this section abbreviate the type-level list representation with promoted-list syntax, such as `'[GPIO 7 C]`, rather than the `Cons (GPIO 7 C) Nil` representation used by the current MicroHs implementation. They also write `Setup` computations with ordinary `do` notation, whereas the implementation uses `QualifiedDo` and `!x.do` because `Setup` is an indexed monad.

```

2  uart <- get_console      -- ns = [],    s = [UART, Unlocked]
3  tzsc_release_periph uart -- ns = [UART], s = [Unlocked]
4  lock_configuration      -- ns = [UART], s = [Locked]

```

The capability ledgers are used to enforce that certain things happens only when they are allowed. For instance, MicroHasTEE requires that the UART is configured *while it belongs to the Secure domain*. Between line 3 and 5 in the `configure` function MicroHasTEE allows the UART to be configured. The program below is rejected by the compiler as ill-formed, as the UART is configured *after* it has been released to the Non-secure domain.

```

1  configure :: Setup '[Unlocked]' '[UART]' '[Locked]' ()
2  configure = do
3    uart <- get_console
4
5    tzsc_release_periph uart
6    uart_init uart $ UARTConfig { baudrate    = 115200
7                                , word_length = 8
8                                , stop_bits   = 1
9                                , parity      = NONE
10                               }
11
12    lock_configuration

```

`uart_init` requires both `Unlocked` and `UART` to be present in the Secure capability ledger. At this point, the Secure ledger is `'[Unlocked]`. Configuration remains open, but the UART is no longer available to Secure code, so the second constraint cannot be satisfied.

Acquiring a resource requires it to be absent from both ledgers. After `tzsc_release_periph`, `UART` is absent from the Secure ledger but remains present in the Non-secure ledger. The second `get_console` call is therefore rejected by the compiler.

```

1  configure :: Setup '[Unlocked]' '[UART]' '[UART, Unlocked]' ()
2  configure = do
3    uart <- get_console
4    tzsc_release_periph uart
5    uart <- get_console

```

The restrictions demonstrated so far ensure that setup maintains a consistent account of what the security attribution on the hardware looks like. The ledger stops representing the attribution of the program if the user can change it *after* we have install callbacks or invoked the application. MicroHasTEE therefore provides a one-way transition that finalizes the capability ledger, disallowing any future modification.

```

1  lock_configuration :: ( Member Unlocked s
2                        , Delete Unlocked s s' )
3                    => Setup ns s ns (Locked ': s') ()

```

The type of `lock_configuration` says that if the Secure ledger contained the element `Unlocked`, the Secure ledger is transformed to one that doesn't contain it, and instead contains `Locked`. All operations that configure the device and change the ledger are gated on the presence of the `Unlocked` element in the Secure capability ledger. Furthermore, the operations that install callbacks or denote Secure functions as callable from the Non-secure domain are similarly gated, and require a program where further configuration has

been ruled out. This act of locking the configuration phase is purely a type-level transition, and not something represented by a corresponding hardware action. The program below is rejected by the compiler, as the red LED is requested after `lock_configuration` was invoked.

```

1  bad = do
2    lock_configuration
3    red <- get_gpio @2 @G -- rejected

```

Finalizing setup produces the ledgers against which the two participant computations are checked. MicroHasTEE represents these computations with the `Secure` and `Nonsecure` monads.

```

1  data Secure    effects a
2  data Nonsecure effects a

```

Peripheral operations are overloaded across these monads. For example, `gpio_toggle` is a method of the `GPIOActions` class.

```

1  class GPIOActions m where
2    gpio_toggle :: Member (GPIO pin port) effects
3                => GPIO pin port -> m effects ()

```

MicroHasTEE provides `GPIOActions` instances for both `Secure` and `Nonsecure`. In either instance, the `Member` constraint requires the GPIO pin to occur in the capability ledger attached to the selected monad. The same operation can therefore be used by either participant, but only when that participant has authority over the pin. Consider the two programs below

```

1  goodNonsecure :: GPIO 7 C -> Nonsecure '[GPIO 7 C] ()
2  goodNonsecure pin = gpio_toggle pin
3
4  badSecure :: GPIO 7 C -> Secure '[Locked, GPIO 2 G] ()
5  badSecure pin = gpio_toggle pin

```

The compiler accepts `goodNonsecure` because its Non-secure ledger contains `GPIO 7 C`. It rejects `badSecure` because its Secure ledger contains only `Locked` and `GPIO 2 G`, not `GPIO 7 C`.

With the underlying mechanisms MicroHasTEE uses to enforce correct security attribution explained, we next give an example of how to install a callback. If the user has properly configured and attributed a GPIO device and its corresponding EXTI device, a callback can be installed against it. When the underlying GPIO pin receives an edge transition, the installed callback is invoked. Consider the function for installing a Non-secure callback below.

```

1  exti_on_nonsecure :: (Member Locked s, Member (EXTI pin port) ns)
2                    => EXTI pin port -> EXTIEdge -> (EXTIEdge -> Nonsecure s)
3                    -> Setup ns s ns s ()

```

The type requires that in order to call this function, attribution must have been finalised (as evident by the `Locked` being present in the Secure capability ledger), and the interrupt belongs to the Non-secure participant. Furthermore, the callback itself is a computation in the `Nonsecure` monad, whose list of effects unifies against the Non-secure capability ledger of the `Setup` monad. This is a good point to reiterate why we lock down the configuration. If unification is successfully done here against the capability ledger of the Non-secure domain, we don't want the program to proceed and make further configurations. This will alter the capability ledger, making the guarantees provided by the prior unification moot.

So far we have described how security attribution is configured, and how callbacks are installed. Crucially, we have not yet shown how the Non-secure domain can invoke secure services from the Secure domain. This is done by explicitly registering *Secure* functions as *callable*, and then providing a carefully typed interface for invoking them. Consider the simple function below.

```
1 add :: Int -> Int -> Secure effects Int
2 add x y = return (x + y)
```

The function `add` takes two parameters and computes their sum in the *Secure* monad. This function on its own cannot be invoked from the *Nonsecure* monad. However, by marking it as *Callable*, we can.

```
1 callable :: (Member Locked s, NonSecureCallable s f)
2           => f -> Setup ns s ns s (Callable f)
3
4 (<.>) :: NFData a => Callable (a -> b) -> a -> Callable b
```

The *Member Locked s* constraint permits a Secure service to be registered only after setup has been finalized. The *NonSecureCallable s f* constraint accepts a Secure computation over the effect ledger *s*, or a function of arbitrary arity that eventually produces such a computation. It therefore unifies the service's effect ledger with the final Secure ledger carried by the surrounding *Setup* computation. For programs expressed through the public interface, a registered service can consequently use only the resources retained by the Secure participant.

The *callable* function returns an opaque *Callable* handle through which the Non-secure participant may refer to the registered service. The *<.>* operator applies one argument to such a handle while preserving the remaining function type.

```
1 f <- callable add
2 let g = f <.> 10 <.> 5
```

The type of *f* is *Callable (Int -> Int -> Secure effects Int)*, while the type of *g* is *Callable (Secure effects Int)*. The type of *g* shows that both arguments have been supplied and that the service is ready to be invoked with *sg*.

```
1 sg :: Callable (Secure s a) -> Nonsecure ns a
```

Given a fully applied handle, *sg* invokes the corresponding Secure computation and returns its result in the *Nonsecure* monad. The function *sg* has quite a straight-forward type. Within Micro-HasTEE's public Haskell interface, *sg* is the only operation that transfers explicit argument and result values between the participants. When the call is made the parameters are passed from the Non-secure domain to the Secure domain, and when the result is returned it is passed from the Secure domain to the Non-secure domain.

Finally, when the callbacks have been installed and the Secure services have been made callable, we invoke the Non-secure application. This is done through the *nonsecure* function. **I should see if I cannot change the code to allow invocation of nonsecure to happen only once...**

```
1 nonsecure :: Member Locked s
2           => Nonsecure ns () -> Setup ns s ns s ()
```

The type states that if the attribution phase has been locked down, we can run the Non-secure application.

A small example of a complete application is shown below.

```
1 {-# LANGUAGE DataKinds #-}
2 {-# LANGUAGE TypeApplications #-}
3
4 module BlinkExample where
5
6 import MicroHasTEE
7
8 type NSEffects = '[GPIO 7 C]
9 type SEffects = '[Locked, GPIO 2 G]
10
11 toggleLED :: GPIO 2 G -> Secure SEffects ()
12 toggleLED = gpio_toggle
13
14 loop :: GPIO 7 C
15       -> Callable (Secure SEffects ())
16       -> Nonsecure NSEffects ()
17 f loop green toggle = do
18   gpio_toggle green
19   sg toggle
20   systick_delay_ms 500
21   loop green toggle
22
23 app :: Setup '[] '[Unlocked] NSEffects SEffects ()
24 app = do
25   board_init
26   board_configure_pll
27   hz <- board_sysclk_hz
28   systick_configure (hz `div` 1000)
29
30   rcc <- get_rcc
31   rcc_enable rcc RCC_GPIOG
32   rcc_enable rcc RCC_GPIOC
33
34   let output = GPIOConfig { mode = OUTPUT
35                             , pull = NOPULL
36                             , alternate = AF0
37                             }
38
39   red <- get_gpio @2 @G
40   gpio_init red output
41
42   green <- get_gpio @7 @C
43   gpio_init green output
44   gpio_release green
45
46   lock_configuration
47   toggle <- callable (toggleLED red)
48   nonsecure (loop green toggle)
49
50 main :: IO ()
51 main = runSetup app
```

The Non-secure loop toggles the green LED directly and toggles the red LED by invoking the registered Secure function through

sg. The red LED belongs to the Secure domain, and can only be manipulated by the Secure application.

MicroHs generates one C source file from each of the two builds. The build system compiles and links each generated file with the MicroHs runtime, startup code, C glue code, and the drivers for its domain to produce the corresponding firmware image.

For programs expressed through its public interface, MicroHasTEE prevents duplicate acquisition, use without ledger authority, configuration after finalization, callbacks attached to the wrong participant, and calls to Secure functions that were not explicitly registered. These guarantees rely on the MicroHasTEE implementation, its hardware backend, and the deployment of matching firmware images. Raw C, FFI, internal interfaces, and malformed gateway input remain outside the static model.

5 Implementation

This section describes how MicroHasTEE turns a source program into the two firmware images deployed on a microcontroller. Our prototype targets an STM32U5A5ZJQ microcontroller, which combines a Cortex-M33 processor implementing Armv8-M TrustZone with 4 MB of flash and 2.5 MB of SRAM. The firmware runs bare metal, without an operating system or RTOS. The prototype therefore includes the linker scripts, startup code, and board support needed to initialize the device and start the MicroHs runtimes.

5.1 Program Partitioning and Compilation

A MicroHasTEE source program contains computations for both the Secure and Non-secure participants, but the two participants execute in separate firmware images. The build system therefore invokes MicroHs twice on the same source program.

In the Secure build, the preprocessor selects an implementation in which **Secure** computations contain executable **IO** actions and **Nonsecure** computations are represented by placeholders. The Non-secure build selects the complementary implementation: **Nonsecure** computations contain executable actions, while **Secure** computations become placeholders. The public types remain the same in both builds, so each build checks the same setup sequence and the same division of resources and computations.

The two interpretations reside in separate implementation modules. The SECURE build flag determines which module the common MicroHasTEE interface re-exports.

```
1 -- Effectful/Internal/Secure.hs
2 data Secure effects a = Secure (IO a)
3
4 data Nonsecure effects a = NonSecure
5
6 -- Effectful/Internal/NonSecure.hs
7 data Secure effects a = SecureDummy
8
9 data Nonsecure effects a = Nonsecure (IO a)
```

These definitions are alternatives and never coexist in one build. Both modules expose the same abstract types and operations, while giving runtime behavior only to computations belonging to the target domain.

Figure 3 illustrates this process. Each compilation retains the behavior associated with its target domain and erases the behavior

of the other participant. MicroHs then generates one C source file from each build. The firmware build links each generated file with a separate MicroHs runtime, the startup code, the C gateway code, and the drivers compiled for that domain. The result is a Secure and a Non-secure firmware image that are flashed into their respective memory regions.

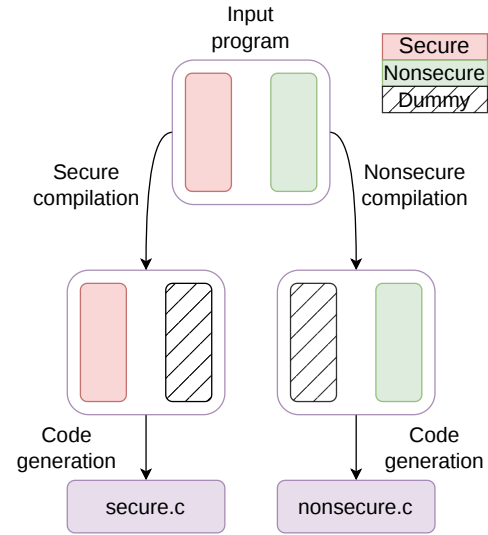


Figure 3: Compilation of one MicroHasTEE program into two domain-specific C programs. The Secure build gives runtime meaning to Secure computations and replaces Non-secure computations with placeholders, while the Non-secure build does the converse. MicroHs generates one C source file for each firmware image.

5.2 Secure Function Registration and Invocation

Although the participant computations have different runtime meanings in the two builds, both builds execute the same sequence of **Setup** operations. The setup state contains the counter used to assign corresponding identifiers to callable functions.

```
1 data Setup nsPre sPre nsPost sPost a =
2   Setup (StateT SetupState IO a)
3
4 data SetupState = SetupState
5   { nonSecureCallable :: [(Int, Ptr BFILE -> IO (Ptr BFILE))]
6   , counter           :: Int
7   , extiCallbacks     :: [ExtiCallback]
8   }
```

Registration of a Secure callable function illustrates why the two executions of setup must remain synchronized. Both builds assign each registration the current counter value and then increment the counter. The Secure implementation uses `mkNSC` to convert the registered function into a dispatch handler, stores the identifier and handler in `nonSecureCallable`, and returns a placeholder handle.

```
1 data Callable a = CallableDummy
```

```

2
3 callable :: (Member Locked s, NonSecureCallable s a)
4   => a -> Setup ns s ns s (Callable a)
5 callable f = do
6   let g inBf = mkNSC f inBf
7   modify $ \st ->
8     st { counter = counter st + 1
9         , nonSecureCallable = (counter st, g) : nonSecureCallable st
10       }
11   return CallableDummy

```

The returned `Callable` value is a placeholder because the Secure firmware does not use the client-side handle. In this build, the Non-secure computation that applies the handle and invokes `sg` is itself represented by a placeholder and is not executed. On the Non-secure side, however, these operations carry a different meaning.

```

1 data Callable a = Callable Int [Ptr BFILE -> IO ()]
2
3 callable :: (Member Locked s, NonSecureCallable s a)
4   => a -> Setup ns s ns s (Callable a)
5 callable _ = do
6   setupst <- get
7   put $ setupst { counter = counter setupst + 1 }
8   return $ Callable (counter setupst) []

```

In the Non-secure build, callable discards the Secure function because Secure computations have no runtime representation in that build. It returns a handle containing the current numeric identifier and an initially empty list of argument serializers. The correspondence between this identifier and the Secure dispatch table depends on both firmware images executing registrations in the same order. MicroHasTEE therefore assumes that matching Secure and Non-secure images are built and deployed together.

After the Secure execution of setup completes, `runSetup` extracts the dispatch table from `SetupState`. Gateway calls may occur after the setup computation has returned, so the implementation creates a `StablePtr` that keeps the table reachable by the Haskell runtime. It stores the resulting opaque token in a Secure C global and retains it for the lifetime of the firmware.

Applying an argument to a callable function appends a writer action to the handle. Given a pointer to a memory-backed BFILE, this action forces the argument to normal form and serializes it into the buffer. Forcing the argument ensures that its evaluation occurs in the Non-secure state before serialization.

```

1 (<.>) :: NFData a => Callable (a -> b) -> a -> Callable b
2 Callable fun writes <.> x =
3   Callable fun (writes ++ [\bf ->
4     x `deepseq` primHSerialize bf x])

```

The MicroHs runtime provides serialization and deserialization primitives for supported Haskell values, so applications do not require hand-written serializers for each callable function. MicroHasTEE uses these primitives to encode a call as a function identifier followed by its arguments. Conceptually, the request and response that passes between the two domains have this format (with `||` denoting concatenation of byte strings).

```

request = serialize(identifier)
        || serialize(arg1)

```

```

|| ...
|| serialize(argN)

```

```
response = serialize(result)
```

The MicroHs runtime format is a combinator graph, and serialised values can be quite large. That is the chief reason for ensuring their evaluation before serialization, through the `NFData` type class.

The Non-secure implementation of the function `sg` produces this serialized package. The code below shows the general operation of this function, but omits details concerning resource cleanup and error handling. Additionally, it is using an internal function `nonsecureLiftIO` to lift this underlying `IO` action to the `Nonsecure` monad, but this function (or any `MonadIO` instance) is not exposed to the end user.

```

1 sg :: Callable (Secure s a) -> Nonsecure ns a
2 sg (Callable fun writeArgs) = nonsecureLiftIO $
3   alloca $ \outLen ->
4   allocaBytes resultBufferSize $ \outBuf -> do
5     request <- c_openb_wr_mem -- fresh, Ptr BFILE
6
7     -- create serialized package
8     primHSerialize request fun
9     mapM_ ($ request) writeArgs
10
11     -- invoke the underlying NSC function
12     c_sg request outBuf resultBufferSize outLen
13
14     -- fetch the result
15     len <- peek outLen
16     response <- c_openb_rd_mem outBuf len
17     primHDeserialize response

```

The function `c_sg` is a Haskell FFI binding to the C function `sg`, which the Secure image exposes through a gateway veneer in the Non-secure Callable region. This is the prototype's only NSC entry point and the only direct cross-domain call path exposed by MicroHasTEE.

The Secure application performs the necessary validation checks, using `cmse_check_address_range` to validate the request metadata and bytes as readable Non-secure memory and the result buffer and length field as writable Non-secure memory. If all checks succeed, it immediately invokes the Secure Haskell runtime by calling a foreign exported Haskell function. The below example shows this Haskell function, but omits details concerning error handling and resource management.

```

1 handle_nsc_call :: Ptr Word8 -> CInt -> Ptr Word8 -> CInt -> Ptr CInt ->
2 handle_nsc_call inBuf inLen outBuf outCapacity outLenPtr = do
3   -- wrap the input bytes in a read-BFILE
4   input <- c_openb_rd_mem inBuf (fromIntegral inLen)
5
6   -- deserialise the function index and look it up in the vtable
7   funId <- primHDeserialize input :: IO Int
8
9   table <- getVTable
10  let fun = lookupFun funId table
11  result <- fun input

```



```

12
13 (resultBuf, resultLen) <- bfileBytes result
14 if resultLen > fromIntegral outCapacity
15 then poke outLenPtr (-1)
16 else do
17   copyBytes outBuf resultBuf resultLen
18   poke outLenPtr (fromIntegral resultLen)

```

The `getVTable` operation reads the token from the Secure C global, reconstructs the `StablePtr`, and dereferences it to recover the dispatch table. The receiver then uses the deserialized function identifier to select a handler from this table.

Every entry in the dispatch table has the uniform handler type `Ptr BFILE -> IO (Ptr BFILE)`, although registered Secure functions may have different arities. The `NonSecureCallable` instances construct this handler recursively from the function's type. For a function of type `a -> b`, `mkNSC` deserializes one value of type `a`, applies the function to it, and continues with the resulting value of type `b`. When the recursion reaches a `Secure` effects `r` computation, it executes the computation and serializes its result. The type-class constraints determine the argument and result types during compilation, while the resulting dispatch table has a uniform runtime representation.

5.3 Peripheral Drivers & Building the Firmware

The prototype MicroHasTEE implementation doesn't use an RTOS, and runs bare-metal. The authors of this paper have hand-written a library of peripheral drivers. The design of the driver library is similar to that of MicroHasTEE, in that they are compiled twice, once for each domain, with the APIs being identical. Some operations are guarded by CPP macros to only be included in the Secure build, such as operations for configuring Security attribution. The compilation mechanism for the peripheral drivers is illustrated in fig. 4.

MicroHasTEE's typed HAL operations invoke this C driver library through the Haskell foreign-function interface. Both builds type-check the same setup operations, but only the Secure driver implementation performs privileged attribution writes to mechanisms such as the TZSC, GPIO security registers, EXTI, and the NVIC. The Non-secure archive retains the same function symbols while compiling privileged operations to no-ops. The shared API therefore supports both interpretations of the setup program without allowing the Non-secure firmware to perform Secure configuration.

Before the application can be flashed to the MCU, the complete firmware must be built. First, the Secure image is built, producing a library file that the Non-secure image must link against to find the NSC entry points. The Non-secure image is then built. Figure 5 shows an overview of the components that go into each of the firmware images, and how they are ordered.

6 Evaluation

MicroHasTEE is a prototype implementation, and we evaluate it primarily through a case study. The purpose of the case study is to establish feasibility of the framework itself, and to illustrate how the security configuration is enforced. After we have presented the case study, we establish the trusted computing base,

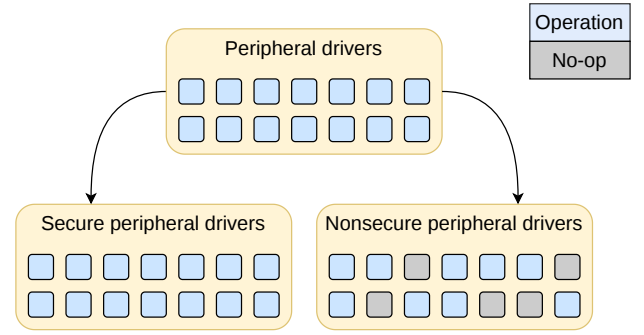


Figure 4: The peripheral driver library split into its Secure and Non-secure builds; some operations become no-ops in the Non-secure build but remain exported to keep a unified API.

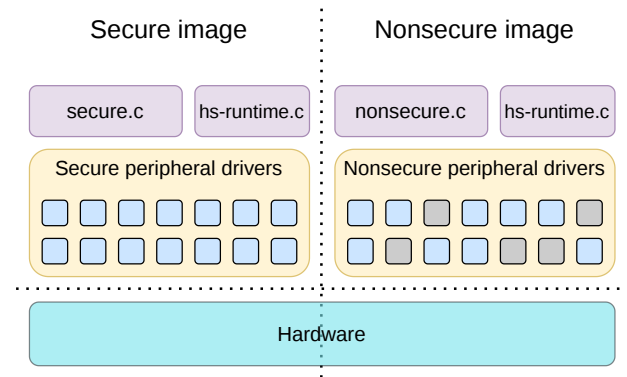


Figure 5: The Secure and Non-secure firmware images, each built from its generated MicroHasTEE C file and the MicroHs runtime on top of that domain's peripheral drivers, running on the shared hardware. The CMSE import library that the Secure build produces is omitted, as it is an artifact used by the linker, and is not present at runtime.

analyse the case study's resource requirements (flash and SRAM), and present some initial performance evaluations.

6.1 Case Study: A Secure Door Lock

blabla describe the application

6.2 Trusted Computing Base

The Trusted Computing Base (TCB) is the software that a programmer needs to trust to implement the Security guarantees(?). It is desirable that this be as small as possible, as more code typically means more bugs. The TCB of MicroHasTEE depends on not just the MicroHasTEE-specific software, but also on the arm-none-eabi-gcc compiler, libc, linker, and assembler. However, for the sake of presentation we omit them and report only the project-specific additions to the TCB that MicroHasTEE adds. The contents of the TCB are listed in table 1.

Table 1: Source-code size of the components in MicroHasTEE's trusted computing base.

Component	SLoC
Peripheral driver library	1588
Startup code	360
Linker scripts	192
MicroHs compiler	10137
MicroHs runtime system	8644
MicroHs base-library source closure (conservative)	13258
MicroHasTEE Haskell Code	2068
MicroHasTEE C code	340
Total	36587

Of the peripheral drivers, just under 500 SLoC are header definitions whereas the remaining 1k SLoC is actual source code. The MicroHs base-library source closure is quite large, at just over 13k SLoC. This is a conservative estimate, however, and the final dependency is most likely much lower. This is the sum of lines of code in the transitive closure of all base-library modules that are loaded during compilation of MicroHasTEE. However, even if we end up using just one or two definitions from a module, its entire content is counted in the reported number. The compiler performs pruning before generating code.

We observe that the code produced for MicroHasTEE consists of 4548 SLoC, while the MicroHs compiler and runtime it relies on contributes the rest. Out of the 2068 SLoC of Haskell code, roughly 1500 SLoC are bindings for the underlying peripheral drivers, and then the effect-tracking wrapper around them. The actual MicroHasTEE multiparty programming framework is surprisingly small, consisting of 400 SLoC.

6.3 Memory Requirements

The MicroHs runtime use graph reduction to execute a program. When a program is executed the runtime system first constructs a graph in memory from a serialized representation emitted by the MicroHs compiler. This graph contains both the programs code and its data, and is modified repeatedly during execution by a byte code interpreter. The graph is updated in place to enable sharing.

When the runtime is compiled for the MCU in this paper, it deliberately omits certain functionality. As an example, the runtime's support for accepting the initial serialized program in a compressed format is not included, as it is not a feature we need. The runtime system itself requires 76 KiB of flash memory, and is roughly the same on both images, totalling 152 KiB.

Additionally, the serialized version of the initial program exists in a `const` array and is similarly compiled and stored in flash. The example program used in the case study requires 67.9 KiB on the Secure image, and 67 KiB in the Non-secure image.

The peripheral drivers occupy much less, but also show more of a difference. This is due to how they are designed, where a noticeable amount of peripheral operations (configuration and security attribution) are only carried out by the Secure image. This has the

Table 2: Flash footprint of the Secure and Non-secure images of the MicroHasTEE door-lock example (STM32U5), attributed per component via the linker map. The C library row is code pulled in transitively from the toolchain (libc/libgcc) rather than written for this project.

Component	Secure (KiB)	Non-secure (KiB)
MicroHs runtime	75.9	75.9
C library (libc/libgcc)	72.3	72.0
Example program	67.9	67.0
Peripheral drivers	7.6	5.1
udb (flash KV store)	5.0	5.0
Bootloader / TZ init	2.6	2.2
Board glue	1.5	1.3
Total	232.7	228.4

effect of the Non-secure peripheral library occupying 2.5 KiB less flash.

How much SRAM is required is more difficult to assess. This stems from the fact that memory usage is difficult to predict in a lazy functional language.

The graph that MicroHs uses is built up of uniformly sized *nodes*. On a 32-bit build that we use for MCUs, a node occupies 8 bytes. The runtime heap-allocates cells in which nodes live, as well as a stack of node pointers. Each stack entry occupies 4 bytes.

While it is somewhat application specific how many heap cells your program will require, for the door-lock case study used in this paper, we allocate 24000 heap cells and a stack with room for 5000 node pointers. We allocate 750 words for the garbage collector (3 KiB), and require an additional 10 KiB for other static globals declared in the `.data` and `.bss` sections.

In total, one firmware image will reliably require *at least* 220 KiB of SRAM at runtime. The MicroHs runtime will not try to allocate more heap cells if the program runs out of free cells, and crashes. The Secure and Non-secure images both require this amount of memory, so across both domains our application requires 440 KiB of SRAM. The STM32U5A5ZJQ we use has 2.5 MiB of SRAM, well beneath our requirements.

6.4 NSC Latency

How long does a NSC call take? Compare with a native C call, to measure the cost.

7 Related Work

MicroHasTEE is inspired by HasTEE, a Haskell framework for confidential computing on Intel SGX. The multiparty programming model and implementation is inspired by HasTEE, but whereas MicroHasTEE is executing in TrustZone-M on an MCU, HasTEE executed in Intel SGX enclaves on larger machines, such as laptops. Furthermore, to run Haskell in the enclave HasTEE used a patched version of the GHC runtime system. MicroHasTEE is compiling and running the upstream version of the MicroHs runtime system and deploys that in TrustZone-M.

rust for TrustZone-M.

There are relatively few projects where a high-level managed language like Haskell is used to run code in TrustZone-M. If we instead divert our eyes to TrustZone-A or Intel SGX, we find more projects.

GoTEECite is a project where a Go compiler and runtime was heavily modified to support *secure routines*, which are executed inside enclaves. Furthermore, GoTEE implements channels for communication between the enclave and the program running outside. The Go compiler was extended with support for automatically partitioning a program, but does otherwise not help the programmer control how sensitive data moves within a program.

JE implements a subset of Java, and comes with support for running in enclaves. It uses Information-Flow Control to ensure that data does not flow through the program in undesirable ways. Compilation is advanced, with a program first undergoing partitioning before another pass ensures that the two programs interactions are not ill-formed.

Concerning running functional languages on MCUs, but not necessarily in a trusted execution environment, a noteworthy project is the *lbm* interpreter. *lbm* is a Lisp interpreter that has been written solely with MCUs being the intended execution environment. It optimises for performance in a resource constrained environment, and implements many features and extensions not usually present in Lisp. It has seen great adoption by the VESC community, where it is used to program motorcontrollers and to implement UIs.

8 Discussion

The project HasTEE did not use MicroHs, and instead used the de-facto Haskell compiler, GHC. They targeted a more conventional platform than a MCU, such as a laptop. Its TCB was, in effect, significantly larger. The GHC runtime system alone exceeds 50000 SLoC.

9 Conclusions & Future Work

MicroHasTEE can reliably reject a whole class of attribution-related bugs, turning runtime errors into compile-time errors. We have shown that a high-level functional language can be used to program resource-constrained MCUs, as well as running it in TrustZone-M.

Evaluation *yet to be done* shows that there is a significant performance cost associated with using MicroHasTEE. The prototype is compiled with a non-optimising compiler, and deployed to a runtime that uses a simple but small bytecode interpreter. For these reasons we are not surprised by the results. A more mature implementation of MicroHasTEE should strive to use a more efficient runtime, but none exists today.

Trying to optimise a program for memory usage is a task not seldomly undertaken by embedded developers. We won't say too much about this, but point out one apple that hangs particularly low. The compiled program is written to flash in a buffer, in a serialized format. The program begins execution by parsing this format and building the program graph in SRAM. Once this initial graph is constructed, the buffer from which it was built is no longer needed.

In our case study, this buffer accounted for almost a third of all the flash necessary. Rather than writing the serialized format to flash, we could instead send it to the runtime system via UART.

We would then not require an in-flash copy of the program, and would save almost 70 KiB.

Acknowledgments

I would like to thank large-language-Joel for always answering any embedded question I have, and Lennart Augustsson.

References

- [1] Lennart Augustsson. 2024. MicroHs: A Small Compiler for Haskell. In *Proceedings of the 17th ACM SIGPLAN International Haskell Symposium, Haskell 2024, Milan, Italy, September 6-7, 2024*, Niki Vazou and J. Garrett Morris (Eds.). ACM, 120–124. doi:10.1145/3677999.3678280

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